

Figure 1 consists of two parts. Part (a) is a T-s diagram showing the thermodynamic cycle. The vertical axis is temperature (T) and the horizontal axis is entropy (s). The cycle includes states 1, 2, 3, 4, 5, and 6. Heat flows are indicated by arrows: Q_{12} (heat input), Q_{23} (heat input), Q_{34} (heat input), Q_{45} (heat input), Q_{56} (heat input), Q_{61} (heat input), Q_{12} (heat input), Q_{23} (heat input), Q_{34} (heat input), Q_{45} (heat input), Q_{56} (heat input), Q_{61} (heat input). Part (b) is a schematic diagram of the energy conversion system. It shows a compressor (压气机) and a turbine (涡轮机) connected by a shaft. A heat exchanger (换热器) is located between them. The heat exchanger has two ports: 1 (heat input) and 2 (heat output). The turbine has two ports: 3 (heat input) and 4 (heat output). The compressor has two ports: 5 (heat input) and 6 (heat output). The heat exchanger is labeled with 1 and 2. The turbine is labeled with 3 and 4. The compressor is labeled with 5 and 6. The shaft is labeled with 7 and 8. The heat exchanger is labeled with 9 and 10. The turbine is labeled with 11 and 12. The compressor is labeled with 13 and 14. The shaft is labeled with 15 and 16. The heat exchanger is labeled with 17 and 18. The turbine is labeled with 19 and 20. The compressor is labeled with 21 and 22. The shaft is labeled with 23 and 24. The heat exchanger is labeled with 25 and 26. The turbine is labeled with 27 and 28. The compressor is labeled with 29 and 30. The shaft is labeled with 31 and 32. The heat exchanger is labeled with 33 and 34. The turbine is labeled with 35 and 36. The compressor is labeled with 37 and 38. The shaft is labeled with 39 and 40. The heat exchanger is labeled with 41 and 42. The turbine is labeled with 43 and 44. The compressor is labeled with 45 and 46. The shaft is labeled with 47 and 48. The heat exchanger is labeled with 49 and 50. The turbine is labeled with 51 and 52. The compressor is labeled with 53 and 54. The shaft is labeled with 55 and 56. The heat exchanger is labeled with 57 and 58. The turbine is labeled with 59 and 60. The compressor is labeled with 61 and 62. The shaft is labeled with 63 and 64. The heat exchanger is labeled with 65 and 66. The turbine is labeled with 67 and 68. The compressor is labeled with 69 and 70. The shaft is labeled with 71 and 72. The heat exchanger is labeled with 73 and 74. The turbine is labeled with 75 and 76. The compressor is labeled with 77 and 78. The shaft is labeled with 79 and 80. The heat exchanger is labeled with 81 and 82. The turbine is labeled with 83 and 84. The compressor is labeled with 85 and 86. The shaft is labeled with 87 and 88. The heat exchanger is labeled with 89 and 90. The turbine is labeled with 91 and 92. The compressor is labeled with 93 and 94. The shaft is labeled with 95 and 96. The heat exchanger is labeled with 97 and 98. The turbine is labeled with 99 and 100.